Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I

1 Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I

Study focus: Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module three on phase equilibria and in today's class we are going to talk about Gibbs phase rule.

Key details

- We will derive the Gibbs phle and we will discuss phase diagrams specifically phase diagram of a one component system.
- If you remember, P stands for the number of phases, C stands for the number of components, and F stands for the degrees of freedom.
- The Gibbs phase rule is going to give us a relation between the number of phases, the number of components, and the number of degrees of freedom for a system existing at equilibrium.
- F is equal to C minus P plus 2.
- What are the conditions a system needs to satisfy in order to be considered to be at equilibrium?
- Firstly, it should be at thermal equilibrium, meaning that the temperature should be constant throughout the system.
- If there exists a difference in temperature, then.
- The system will try to equalize the temperature and it will reach equilibrium only when the temperature is same throughout the system.
- Talked about a new term chemical potential.
- When we talk of any chemical reaction, we know that gibbs free energy decides.

- the direction of the reaction.
- Wherever there is decrease in free energy, we know that the reaction is going to move in that direction because every system tries to go towards minimum free energy.
- But free energy cannot be used as a criteria for chemical equilibrium because it is.
- An extensive property.
- Therefore, a new term was defined which is known as the chemical potential.
- So for a system to be at chemical or material equilibrium, the condition is that chemical potential of the substance should be same in all the phases.
- Alpha is equal to mu i beta is equal to mu i gamma.
- What does it mean?
- That the chemical potential of the I component in the alpha phase is equal to the chemical potential of the I component in the beta phase, which is again equal to chemical potential of the I component in gamma.
- phase and so on.
- Our system is where all the C components are distributed between P phases.
- If we want to talk about any system at equilibrium, first and foremost, we should know what are the variables we need to ascertain.

Example or application from the lecture

- The phase rule was given by Williams Gibbs in eighteen seventy four.
- Before deriving the Gibbs phase rule, let us understand what we mean by a system at equilibrium.
- molar gives free energy which is chemical potential and since there are so many species in the system we have to consider partial molar gives free energy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 05:00–09:59

Firstly we will have temperature.

Key details

- We should know the temperature, so that should be one variable and then we should know the pressure, so that is again another variable.
- To know and also the pressure of the system which we need to know.
- The composition we need to know the mole fraction of each component in every face.
- For each phase, we know that the sum of the mole fractions is equal to unity, because this is a relative way of expressing composition, mole fraction.
- for all the components in that particular phase.
- Say alpha phase.
- In the alpha phase, we have so many components.
- There are C components.
- Plus mole fraction of the third component of the alpha phase, so on, till we come to the last component that is the mole fraction of the component C in the alpha phase.

- And all this should sum up to one.
- We are interested in the number of components.
- Why C minus 1?
- Because we know that all these will sum up to 1.
- Now coming to the second phase, say that is the beta phase, again we have an equation like this where we have the first component mode fraction, the second component mode fraction, third component mode fraction, so on adding up to one.
- And similarly for the gamma phase, and as we go on, we know that there are p number of phases.
- So therefore for the last phase it will be the mole fraction of the first component in p phase plus the mole fraction of the second component in p phase, so on.
- Up to the last component in the P phase, again this all adds up to one.
- Therefore, we have C minus one compositions again need to be determined.
- For each of the phases, we need C minus 1 more fraction terms which have to be define and since we have p number of phases therefore the number of composition variables will be p times c minus one.
- Plus we have temperature and pressure which we need to ascertain.
- So therefore the total number of intensive variables will become P times C minus one plus two.
- If we want to find out what are these variables.
- We need to know the number of equations or the constraints.
- At equilibrium, we know that chemical potential of a particular component is the same in every phase of the system.
- Just now we define chemical equilibrium and talked about this equation.
- So the first component chemical potential in the alpha phase should be equal to.
- The chemical potential of the first component in the beta phase and also that should be equal to chemical potential of the same component in the gamma phase and so on.
- So the first component chemical potential should be equal in all the p phases.

Example or application from the lecture

- So first let us talk of.
- So the equation would be given by the mole fraction of the first component in the alpha phase plus the mole fraction of the second component in the alpha phase.
- So let us look at it for the first component.

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Core explanation · 10:00–14:59

Second component chemical potential in alpha phase should be equal to the chemical potential of the second component in the beta phase, and also that again equal to the chemical potential of the second component in gamma phase.

Key details

- P because P is the total number of phases, therefore what we get here is again P minus one number of equations.
- Therefore, again we have P minus one number of equations.
- So therefore the total number of equints now we have is C times P minus 1.
- Relating X and Y.
- I cannot change X and Y independently.
- If X changes then Y is fixed if Y changes then X is fixed.
- Because of this one equation the degree of freedom or the.
- My independence to change X and Y is reduced by 1.
- And this is Gibbs phase root.
- We will take up the water system because that is the easiest for us to relate to.
- In that case, C is one because it is a one component system.
- P is one again.
- because we are talking about one phase and when we apply the Gibbs phase rule F is equal to two.
- That temperature and pressure can be varied independently.
- And when we have two phases present in.
- So either the solid is in equilibrium with the liquid or the liquid is in equilibrium with the vapor or the solid is in equilibrium with the vapor, then we know that there are two phases, C is again one because it's a one component system.
- So that means now the system has become a univariant system, and we know that when we have two phases in equilibrium, either the temperature or the pressure can be varied independently, but not both.
- And when we have all the three phases in equilibrium.
- And this system now, what happens?
- F is equal to 0.
- C is equal to 1, P is equal to 3.

Example or application from the lecture

- Since we know the number of variables and the number of equations, the degrees of freedom will be given by the total number of variables minus the number of equations which we have.
- Let us understand this.
- Determine now since there exists no relation, suppose there are independent variables, then that means I need to ascertain both X as well as Y in order to define the system.

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Core explanation · 15:00–19:59

an invariant system.

Key details

- It is a diagram which represents the conditions under which a substance exists in different phases in a system.

- How do I construct a phase diagram of a one component system?
- The phase rule we know which we are going to apply is F is equal to C minus P plus 2.
- If we have only a single phase, we want to represent a single phase in the diagram, we know that using this f is equal to 3 minus p, since it's one phase, f becomes equal to 2.
- And in a diagram, this would be represented by an area.
- When we have three phases in equilibrium then it will be represented by a point, but if you have just a single phase that would be shown by an area in the diagram.
- This area would be the solid phase.
- And in between we have the liquid phase.
- Say this area is going to be the solid phase, this area will be the gaseous phase, and in between we have the liquid phase.
- That would be represented by an area.
- Equilibrium between solid and the liquid.
- And similarly, liquid and vapor, so that would be this line here.
- And then if there is vapor and solid, then this line would represent the.
- And then if we have the solid liquid as well as the vapor present in equilibrium together, then that would be represented by a point, and that's exactly what we see here.

Example or application from the lecture

- Let us now discuss the phase diagram.
- Let us see what the phase diagram is going to look like.
- Suppose we have just one phase, and just now we said that when we have one phase, phase rule tell us that there is going to it is going to be represented by an area on the diagram.

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Core explanation · 20:00–24:59

as well as the vapor in equilibrium and that would be a point on the phram.

Key details

- So in the next class.
- We will take up the water system and see exactly what the phram is going to look like for a water system.

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EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

C Revision and retrieval practice